

Let's have a look at each in detail.

- **SOAP API**

SOAP is a standard protocol defined by the W3C Standard. It is used for sending and receiving API requests and responses. SOAP uses XML format in WSDL (Web Services Description Language) to send and receive data. It uses the simple HTTP protocol. SOAP API's have been decreasing in popularity over the years due to its integration complexities across enterprises.

- **REST API**

REST stands for REpresentational State Transfer. It is an architectural style for distributed hypermedia systems. Roy Fielding first presented REST in 2000. It generally runs over [HTTP](#). There are various HTTP codes that REST uses to convey the results of a client's request. For instance, code 201 and 202 indicate 'created,' and 'accepted' respectively, whereas code 301 indicates 'moved permanently.' REST is usually used as an alternative to SOAP as it uses simple URLs for requests in some cases, instead of XML. REST API supports XML and JSON format. Yet, it typically uses JSON in message bodies over XML. REST is generally used for mobile and web apps.

Now that you have an idea about the API architectural types, here is a look at how to test each using SoapUI.

How to test SOAP API on SoapUI?

Testing SOAP API on SoapUI is a relatively easy task. Here's a step-by-step guide to test SOAP API on SoapUI.

1. Create a SOAP project

Go to the Menu toolbar -> Click on File -> New SOAP project. You can also create a new SOAP project with the Ctrl+N keyboard shortcut quickly.

2. In the pop-up dialogue box that follows, enter the project name. Next, you'll need to insert the 'Initial WSDL.' This can be done in two ways. First, copy-paste an URL from the internet in the tab.

For example, <http://www.dataaccess.com/webservicesserver/numberconversion.wso?WSDL>

Or, you can enter an already downloaded file from your computer by

